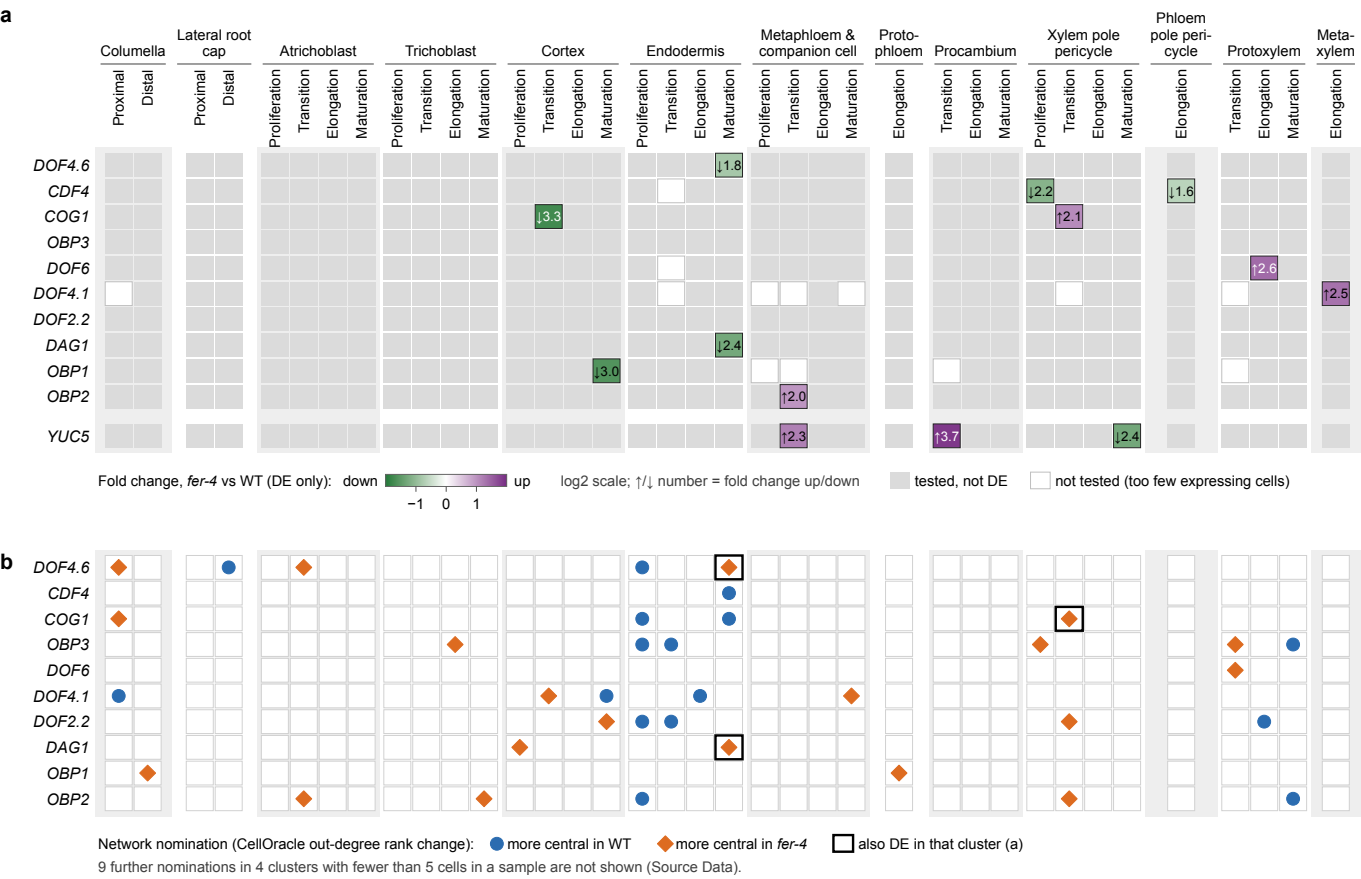

a
Columella
Proximal
Distal
Lateral root cap
Proximal
Distal
Atrichoblast
Proliferation
Transition
Elongation
Maturation
Trichoblast
Proliferation
Transition
Elongation
Maturation
Cortex
Proliferation
Transition
Elongation
Maturation
Endodermis
Proliferation
Transition
Elongation
Maturation
Metaphloem & companion cell
Proliferation
Transition
Elongation
Maturation
Proto-phloem
Elongation
Procambium
Transition
Elongation
Maturation
Xylem pole pericycle
Proliferation
Transition
Elongation
Maturation
Phloem pole peri-cycle
Elongation
Protoxylem
Transition
Elongation
Maturation
Meta-xylem
Elongation
DOF4.6
CDF4
COG1
OBP3
DOF6
DOF4.1
DOF2.2
DAG1
OBP1
OBP2
YUC5
↓1.8
↓2.2
↓1.6
↓3.3
↑2.1
↑2.6
↑2.5
↓2.4
↓3.0
↑2.0
↑2.3
↑3.7
↓2.4
Fold change, fer-4 vs WT (DE only): down
up
−1 0 1
log2 scale; ↑/↓ number = fold change up/down
tested, not DE
not tested (too few expressing cells)
b
DOF4.6
CDF4
COG1
OBP3
DOF6
DOF4.1
DOF2.2
DAG1
OBP1
OBP2
Network nomination (CellOracle out-degree rank change):
more central in WT
more central in fer-4
also DE in that cluster (a)
9 further nominations in 4 clusters with fewer than 5 cells in a sample are not shown (Source Data).